A

PROJECT REPORT

ON

VidSnapAI

Submitted to
G H Raisoni University, Amravati.
School of Science

In partial fulfillment for the requirement of the degree of

Bachelor of Computer Application
BCA III Year, Semester V

Submitted by

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BCA III Year, Semester V

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Department of Computer Applications
School of Science

2025-26

G H Raisoni University, Amravati
Department of Computer Applications
School of Science
2025-26

CERTIFICATE

This is to certify that the project titled “VidSnapAI” has been submitted by Piyush Ramteke in partial fulfillment for the requirement of the degree of Bachelor of Computer Application BCA III Year, Semester V. This is an original work carried out by him under the guidance of Mr. Sarang Thakare and he have undergone the requisite hours of project work as specified by G H Raisoni University, Amravati.

Mr. Sarang Thakare
Project Guide

Dr. Bhavana Karmore
Head of the Department

Dr./Mr.……
External Examiner

Declaration

Ihereby certify that the work which is being presented in the project entitled "VidSnapAI - AI-Powered TikTok/Reel Generator" submitted in the DEPARTMENT OF COMPUTER APPLICATIONS AT G H RAISONI UNIVERSITY, AMRAVATI is an authentic record of my own work carried under the supervision of MR. Sarang Thakake & Professor This study has not been submitted to any other institution or university for the award of any other degree.

Yours Faithfully

Piyush Ramteke

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BCA III, Sem V

Date:

G H Raisoni University, Amravati
Department of Computer Application
SchoolOf Science

2025-26

ACKNOWLEDGEMENT

I would like to express my profound gratitude to Dr. Bhavana Karmore, Head of Department of Computer Application, and Dr. Shailesh Thakare Dean of SCHOOL OF SCIENCE , G H RAISONI UNIVERSITY for their contributions to the completion of my project titled VidSnapAI – AI-Powered TikTok/Reel Generator .

I would like to express my special thanks to project guide Prof. Mr. Sarang Thakare for his time and efforts he provided throughout the year. Your useful advice and suggestions were really helpful to me during the project's completion. In this aspect, I am eternally grateful to you.

Iwould like to acknowledge that this project was completed entirely by me and not by someone else.

Piyush Ramteke BCA
III Year, Sem V
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Abstract

VidSnapAI – AI-Powered TikTok/Reel Generator

VidSnapAI is an innovative AI-powered SaaS platform that automates the creation of short-form social media videos such as TikToks and Instagram Reels using user-provided images and text descriptions. The platform simplifies the content creation process by integrating advanced voice synthesis and automated video editing, eliminating the need for manual post-production. Built on a Python Flask backend, VidSnapAI enables users to upload images and text, which are processed through the ElevenLabs API to generate natural-sounding voiceovers using cutting-edge text-to-speech (TTS) technology. The FFmpeg media toolkit then compiles the images into a vertical 9:16 video format, synchronizing them with the synthesized audio to produce a polished, ready-to-share video. The completed videos are displayed in a web-based gallery, allowing users to preview, download, or instantly share their creations. Designed with a mobile-first approach, VidSnapAI ensures seamless compatibility with smartphone screens and major social media platforms. By combining AI-driven audio generation and automated video assembly, the platform offers a fast, user-friendly, and professional-grade solution for digital creators, marketers, and educators seeking efficient and scalable media production workflows.

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Chapter 1: Introduction

In the modern digital era, social media platforms like TikTok, Instagram Reels, and YouTube Shorts have revolutionized the way users create and consume content. Short-form videos have become the most engaging medium for communication, marketing, and entertainment. However, creating such videos manually requires considerable time, technical skill, and creativity. Tasks such as scripting, recording voiceovers, editing visuals, and synchronizing audio are often challenging for non-professional users. To overcome these challenges, VidSnapAI has been developed as an AI-powered system that automates the creation of TikTok and Reel-style videos.

VidSnapAI integrates advanced artificial intelligence tools to simplify video production. The system allows users to provide a short text description or a set of images, which it then transforms into a complete, high-quality video with AI-generated voice narration and synchronized visuals. It uses Flask as the web framework to manage user interactions, ElevenLabs API for realistic text-to-speech voice generation, and FFmpeg for video processing and assembly.

The project's primary objective is to minimize manual editing and provide a seamless, automated solution for quick and professional video generation. By leveraging AI and automation, VidSnapAI empowers creators, educators, and marketers to produce engaging content efficiently. The project demonstrates how combining AI-based voice synthesis and multimedia processing can significantly enhance productivity in the digital content industry. In essence, VidSnapAI bridges the gap between creativity and technology, enabling effortless and intelligent short-video creation for modern social platforms.

1.1 Problem Statement

2In the current digital era, short-form video platforms such as TikTok, Instagram Reels, and YouTube Shorts have become dominant channels for communication, entertainment, and marketing. These platforms demand a continuous flow of creative and engaging video content. However, producing high-quality short videos requires considerable effort, time, and technical expertise. The traditional workflow —comprising scriptwriting, recording voiceovers, editing clips, adding transitions, and synchronizing background music—is both time-consuming and resource-intensive. For individual creators, students, and small businesses, this becomes a major limitation, as they often lack the professional tools or skills to produce visually appealing and well-narrated videos consistently.

3The main problem addressed in this project is the lack of an automated, user-friendly system that can generate short videos efficiently using artificial intelligence. Existing video editing tools require manual operation and do not leverage AI capabilities to automate narration or editing. This results in creative burnout, inconsistent content quality, and delayed publishing schedules.

4 To overcome these challenges, VidSnapAI aims to develop an AI-driven web application that automatically converts user-provided text and images into short, platform-ready videos. By integrating ElevenLabs for realistic AI-based text-to-speech conversion and FFmpeg for automated video compilation, the system eliminates manual editing and recording steps. The project seeks to empower content creators by simplifying video production through automation, enabling faster turnaround, consistent quality, and greater accessibility for non-technical users.

4.1 Objectives

Automation o f Short-Video Creation:
Todevelop an intelligent system that automatically generates TikTok or Instagram Reel-style videos byintegrating user text, audio, and imagesintoa coherent output without manual editing.

Integration o f AI-Based VoiceSynthesis:

Toutilize ElevenLabs Text-to-Speech (TTS)technology for converting user-provided scripts into realistic, human-like audio narration, therebyimproving engagement and content quality.

Efficient V ideo Processing and Compilation:

Toemploy FFmpeg, a robust multimedia processing library, for assembling multiple media elements —such as voiceovers, images, and clips—into high-quality vertical videos optimized for mobile screens.

Development o f aFlask-Based Web Interface:

Todesign a user-friendly web application using Python's Flask framework that allows users to upload inputs, trigger AI generation, and view or download the finished video directly through a browser.

Enhancement o f Productivityand Accessibility:

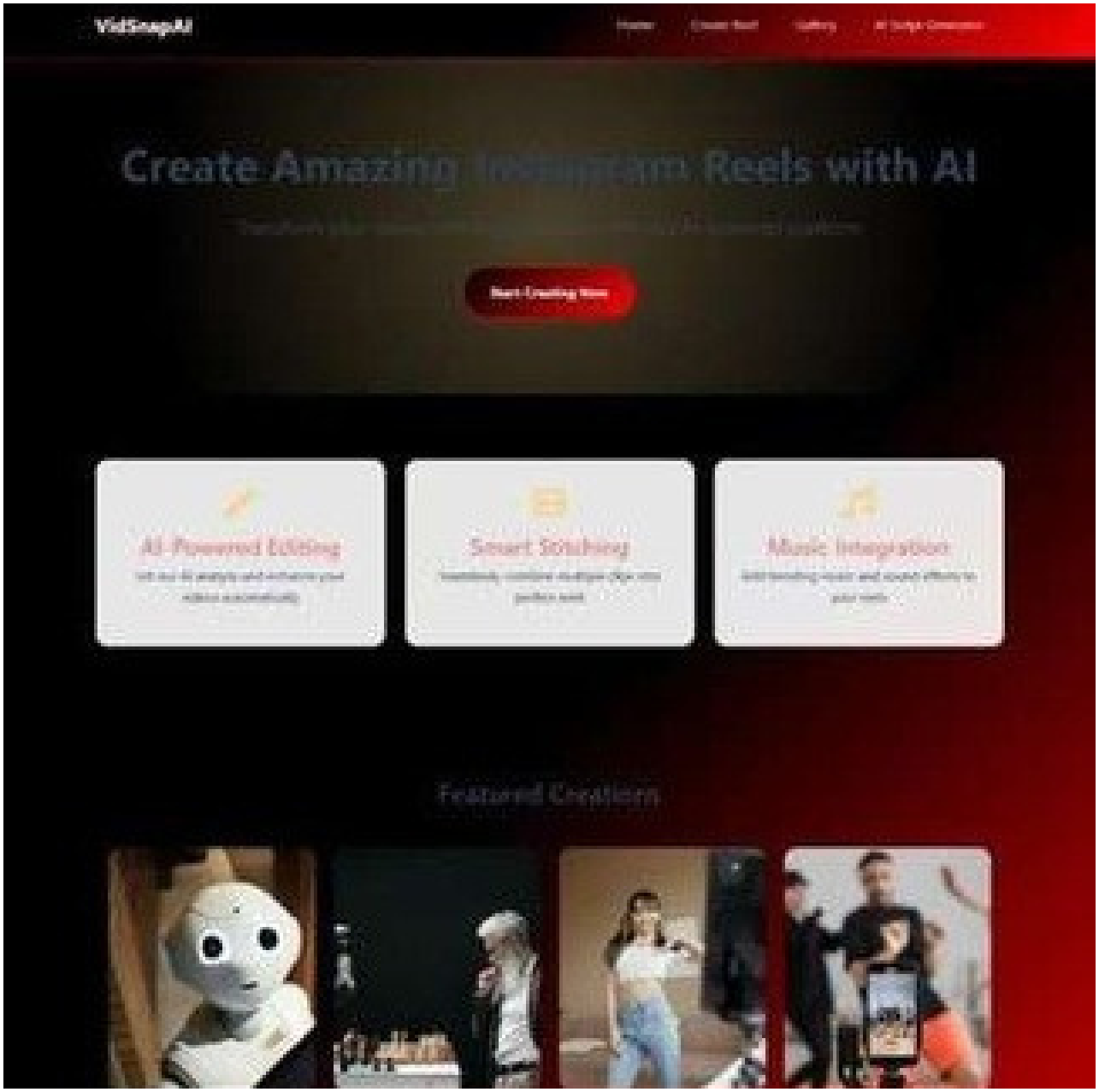
Tominimize the time, cost, and technicalskill required for content production, enabling students, influencers, and small businesses to createprofessional-grade videos efficiently.

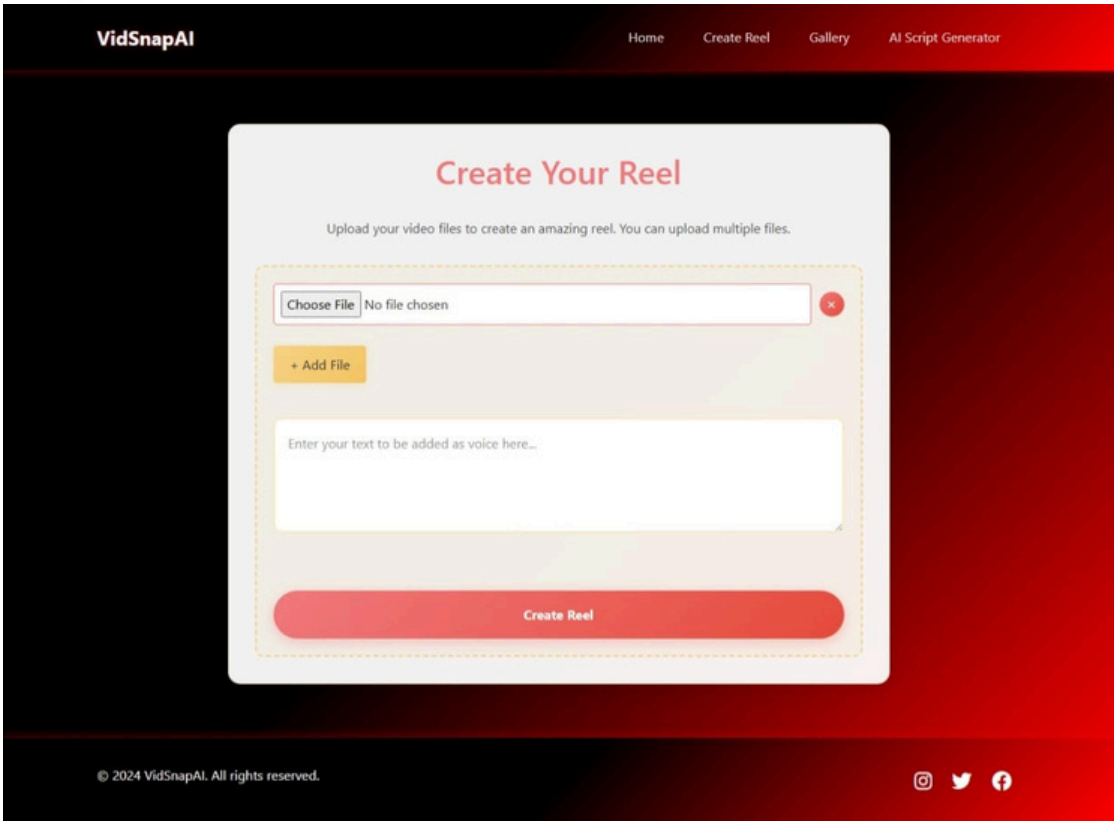
Scalability a nd Future Improvements:

Todesign the project in a modular, scalablemanner that allows future integration of multilingual narration, dynamic video templates, and advanced editing effects.

4.2 Scope

The scope of VidSnapAI covers the automation of TikTok/Reel style video production from start to finish. The system will take user input (text script or media) and produce a finalized video with AI-generated voiceover. It will include core modules for text-to-speech, video assembly, and a user interface. The project focuses on short video creation and does not cover advanced editing features like complex visual effects or multi-language translation beyond English. The intended users are content creators and businesses who need quick video turnaround. Limitations include reliance on the quality of AI tools (voiceover clarity, video assets available) and constraints of running on standard computing hardware. Future extensions could add more languages, templates, or integration with social platform




VidSnapAI
Home
Create Reel
Gallery
AI Script Generator
Create Your Reel
Upload your video files to create an amazing reel. You can upload multiple files.
Choose File
No file chosen
+ Add File
Enter your text to be added as voice here...
Create Reel
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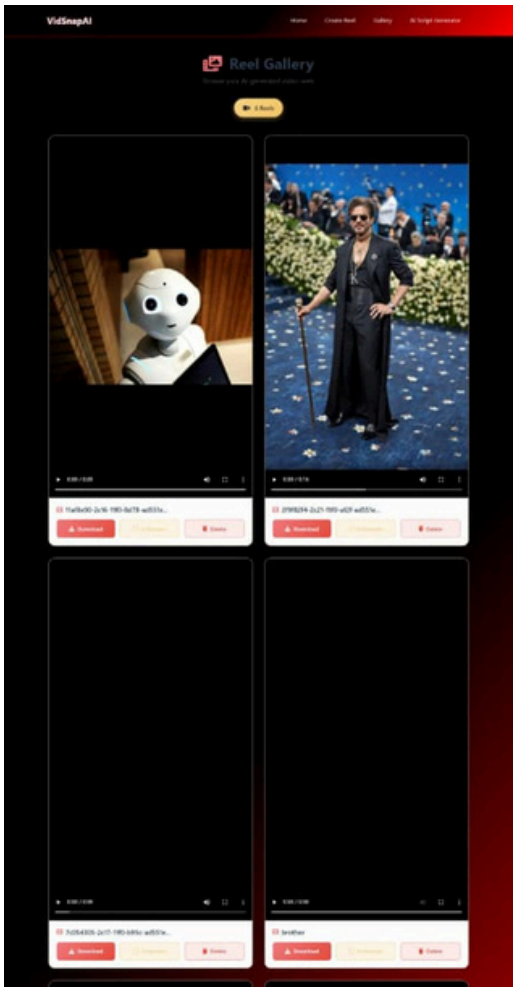
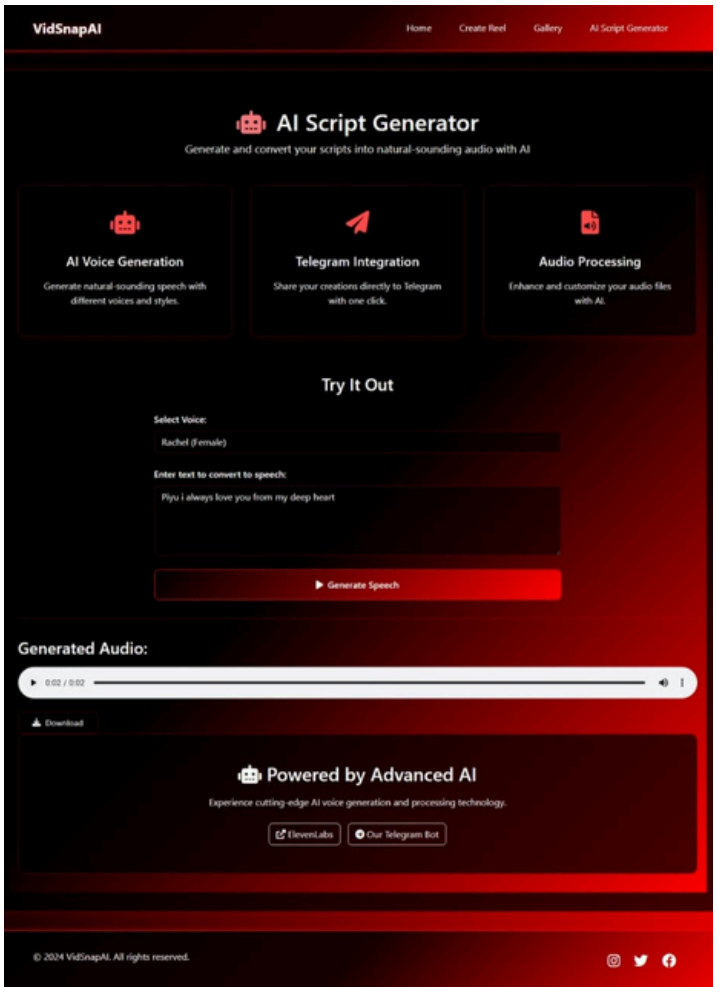

VidSnapAI
Reel Gallery


VidSnapAI
Home
Create Reel
Gallery
AI Script Generator
AI Script Generator
Generate and convert your scripts into natural-sounding audio with AI
AI Voice Generation
Generate natural-sounding speech with different voices and styles.
Telegram Integration
Share your creations directly to Telegram with one click.
Audio Processing
Enhance and customize your audio files with AI.
Try It Out
Select Voice:
Rachel (Female)
Enter text to convert to speech:
Piyu i always love you from my deep heart
Generate Speech
Generated Audio:
Download
Powered by Advanced AI
Experience cutting-edge AI voice generation and processing technology.
ElevenLabs
Our Telegram Bot
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Chapter 2: Literature Review

Thereis growing industryinterest in automating short-video content creation. For example, TikTok's "Smart Split" feature can automatically edit longer footage into shorter clips using AI theverge.com. This illustrates how AI can assist in video editing tasks by identifying key segments. In addition, web- based tools like Revid.ai offer TikTok video generation from text or URLs (without human editing). Notably, Zapier reports that Google's new Veo 3 (Gemini) can create entire high-quality TikTok videos from simple prompts zapier.com, indicating rapid advances in AI-driven video synthesis.
AI voice generation is also a critical component. According to ElevenLabs, their "AI voices

VidSnapAI combines advanced artificial intelligence, automation, and multimedia processing to generate short-form videos efficiently. It utilizes text-to-speech (TTS) technology powered by the ElevenLabs API to convert user-provided text into realistic voiceovers. This innovation is grounded in research on neural speech synthesis, which uses deep learning models to produce human-like speech patterns with natural intonation and emotion. The adoption of such models enhances user engagement by creating authentic-sounding audio narratives that align with the video's visual flow.

For video generation, VidSnapAI employs FFmpeg, a robust open-source media toolkit widely recognized in academic and industrial research for its efficiency in multimedia processing. FFmpeg enables frame-by-frame image compilation, synchronization of audio tracks, and output formatting in vertical 9:16 aspect ratio—ideal for mobile and social media platforms. Previous studies in automated media generation emphasize the effectiveness of such tools in reducing manual workload while maintaining professional output quality.

The integration of Flask, a lightweight Python web framework, supports scalability and user interaction through an intuitive web interface. Literature on web-based AI systems highlights Flask's flexibility in combining APIs, machine learning modules, and user-uploaded data seamlessly. Furthermore, VidSnapAI's mobile-first design aligns with research emphasizing responsive interfaces as a key factor for user engagement and accessibility across devices.

Overall, VidSnapAI reflects current trends in AI-assisted content creation, where automation, accessibility, and personalization converge. By integrating natural language processing, speech synthesis, and multimedia automation, VidSnapAI exemplifies the next generation of smart media production tools. Its development contributes to ongoing research in human-computer interaction, creative automation, and digital media technologies aimed at enhancing productivity for creators, marketers, and educators.

Chapter 3: Methodology

3.1 System Analysis

The VidSnapAI system is designed to automate the creation of short-form social media videos by integrating AI-driven voice synthesis and automated video generation. The system analysis involves understanding the problem, identifying system requirements, and defining how the system achieves its objectives.

Problem Analysis:
Traditional video creation requires significant manual effort — recording audio, editing clips, and synchronizing content. This process is time-consuming and often requires technical expertise. The need for a simple, automated solution led to the development of VidSnapAI, which allows users to create TikTok or Instagram Reels quickly using AI tools.

Existing System:
Existing tools like manual video editors (e.g., Adobe Premiere, CapCut) demand user effort in recording, editing, and synchronizing. These lack automation and AI integration for voice generation.

Proposed System:
The proposed system uses Python Flask as the backend framework, ElevenLabs API for generating realistic AI voiceovers, and FFmpeg for video compilation. The user uploads images and text, and the system automatically generates an AI-narrated reel in vertical format.

Functional Requirements:
- Input text/script and media upload by user.
- Automatic generation of AI voice from text using ElevenLabs.
- Combine audio and images into a vertical video using FFmpeg.
- Display and download the final video from a web interface.

Non-Functional Requirements:
- Usability: Simple and accessible web interface.
- Performance: Efficient video processing and minimal latency.
- Reliability: Error handling for missing or invalid inputs.
- Compatibility: Works across Windows, Linux, or macOS.

3.2 System Design

The System Design phase defines how VidSnapAI components interact to produce the final output efficiently. It involves designing the architecture, data flow, and integration of AI and multimedia tools.

Architecture Overview:
VidSnapAI follows a modular architecture consisting of three main components:

1 .Frontend Interface (User Layer):
- Created using HTML, CSS, and Bootstrap.
- Allows users to upload images and input text descriptions.

2 .Backend Processing (Application Layer):
- Built using Flask, a lightweight Python web framework.
- Manages routes for uploading files, processing requests, and returning results.
- Handles integration with ElevenLabs API for text-to-speech.

3 .Media Generation (Processing Layer):
- Uses FFmpeg for merging images and audio into video format.
- Ensures 9:16 aspect ratio suitable for TikTok/Reels.

Data Flow Diagram (Conceptual):

User Input → Flask Server → ElevenLabs (Voice Generation) → FFmpeg (Video Creation) → Flask (Output Display)

Design Modules:
- Module 1: Input Module – Accepts text and media uploads.
- Module 2: AI Voice Generation Module – Sends text to ElevenLabs API to generate voice narration.
- Module 3: Video Generation Module – Combines visuals and audio into a final reel using FFmpeg commands.
- Module 4: Output Module – Displays generated videos on a web gallery page for viewing or download.

Technology Stack:

Component	Technology Used	Purpose
Backend	Flask (Python)	Handle routes and logic
AI Service	ElevenLabs API	Text-to-Speech voice generation
Video Engine	FFmpeg	Video and audio processing
Interface	HTML, CSS, Bootstrap	User interaction
Storage	Local/Temporary directory	Store media files

System Flow Summary:

- The user uploads media and provides a text script.
- Flask routes the text to ElevenLabs API for voice synthesis.
- The API returns an MP3 audio file.
- FFmpeg merges the audio with user images into an MP4 video.
- The Flask server displays the generated video in a downloadable gallery.

Chapter 4: Proposed System

4.1 Working of System

1. TheVidSnapAI system automates short video creation by combining text input, AI-generated voice, andvideo processing into a single workflow. The working of the system is organizedinto the following steps:

2. User Input Module:
   The process beginswhen theuserenters a text description or short script through the web interface builtusing Flask.Optionally, the user can upload images or select default templatesforthe videobackground.

3. Text-to-Speech C onversion:
   The entered text is processedusingthe ElevenLabs API, which generates a natural and realistic voice narration.Theaudio file (usually in MP3 format) is saved temporarily on the server forthenext stage.

4. Video Generation Pipeline:
   The system uses FFmpeg ,amultimedia framework, to merge the AI-generated voice with selected visuals. It arrangesimages or clips in sequence, synchronizes them with the narration, and outputsashortvideo formatted in a 9:16 ratio suitable for TikTok or Instagram Reels.

5. Automation a nd Integration:
   All tasks are executedthroughasingle Python pipeline, ensuring that voice generation, video creation, and filemanagement are automated without manual editing.

6. Outputa nd Preview:
   Once processing is complete, the generated video is stored on the server and displayed in a gallery interface,allowing the user to preview, download, or share it directly.

   Overall, VidSnapAI provides an efficient end-to-end solution that transforms a simple text prompt into a fully edited, social media-ready short video using artificial intelligence and automation

These steps allow a user to get from a text idea to a shareable TikTok-style video without manual editing. Throughout, the system automatically enforces TikTok's constraints (e.g. aspect ratio, duration). Error checking ensures that if any step fails (e.g. API call or FFmpeg error), the user is notified.

4.2 Hardware and Software Requirements

Hardware Requirements :

1. Processor: Minimum Intel Core i5 (7th Gen or higher) / AMD Ryzen 5 or equivalent to handlesimultaneous audio-video processing efficiently.

2. RAM: At least 8 GB RAM is required for smooth execution of Python scripts and multimedia operations. 16 GB RAM is recommended for faster video generation.

3. Storage: A minimum of 256 GB SSD (or 500 GB HDD) is needed to store temporary audio, video, and output files created during the rendering process.

4. Display: A 1080p monitor with standard color support to preview generated videos accurately.

5. Peripherals: Keyboard, mouse, and stable internet connectivity are mandatory for uploading content, accessing APIs, and downloading generated videos.

Software Requirements :

1. Operating System: Microsoft Windows 10 or 11 (64-bit). The project is fully compatiblewithanyOS supporting Python 3.x.

2. Programming Language: Python 3.8 or later, used for backend logic and integration of APIs.

3. Framework: Flask micro-web framework for developing the web-based user interface and managing HTTP requests.

4. Libraries and Tools:
   - o ElevenLabs API for AI-based text-to-speech voice generation.
   - o FFmpeg for combining audio and image/video frames into final reels.
   - o Requests, dotenv, ffmpeg-python, and pandas libraries for processing and environment management.

5. Text Editor/IDE: VS Code or PyCharm for code editing and debugging.

6. Web Browser: Google Chrome or Edge to test and run the local Flask server interface.

These configurations ensure that VidSnapAI operates efficiently, enabling seamless AI voice generation, video assembly, and real-time preview during short-video production.

Chapter 5: Conclusion

This project demonstrates the feasibility of using AI to automate TikTok/Reel video generation. VidSnapAI successfully integrates ElevenLabs' AI voice synthesis and the FFmpeg media framework to convert a simple text script into a polished short video. The system's Flask interface makes it accessible to users without requiring them to learn complex editing tools. In practice, VidSnapAI can significantly reduce the time and effort needed to produce short videos. During prototype testing, the tool generated sample videos with engaging narration and coherent visuals, validating the design approach.

In conclusion, VidSnapAI meets its objectives by providing a streamlined pipeline for video

creation. It shows that combining existing technologies (ElevenLabs for voice, FFmpeg for editing) can produce a new, useful application. The project highlights how AI can empower content creators, enabling even novice users to produce professional-quality content quickly.

Future improvements could include adding multiple voice languages, expanding the library of video templates, and optimizing processing speed. Overall, VidSnapAI lays the groundwork for arobust content generation platform and offers a proof-of-concept for AI-driven media production.

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FFmpeg – About FFmpeg (Official documentation) ffmpeg.org.

Flask (web framework) – "Flask is a micro web framework written in Python"en.wikipedia.org.

Liszewski, A., The Verge, "TikTok can use AI to turn your long video into short ones"

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[Additional technical documentation and API references for Flask, ElevenLabs, and FFmpeg were consulted.]

Summary

VidSnapAI successfully automates the creation of TikTok/Reel videos by integrating AI-driven voice synthesis and automated editing. The system uses ElevenLabs for natural-sounding narration elevenlabs.io and FFmpeg for versatile multimedia processing ffmpeg.org, controlled via Python. A Flask-based web interface allows users to input content and quickly receive a finished short video, lowering the barrier to content creation. Initial tests show the prototype can rapidly produce engaging videos, confirming the project's viability. Future enhancements could include support for additional languages, a larger selection of video templates, and improved performance tuning. These improvements would make VidSnapAI an even more powerful tool for creators.